

Linear Algebra Mastery

Vectors and Vector Operations

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1 Introduction

A vector is an ordered list of arrays with both magnitude and direction. Vectors are very common in machine learning applications. Whenever a linear regression model is trained, the predictions are presented using vector-linear combination method e.g. $\hat{y} = w_1x_1 + w_2x_2 + \dots + w_nx_n$. There are many other applications of vectors in real world, they are not just limited to linear regression modelling. Some of them includes:

1. Scaling a feature, done in feature normalization
2. Understanding span of all possible predictions a model can make.
3. Compute model weights, from $A\vec{x} = b$, given that $\vec{x}$ is a vector with the model weights.

2 Formal Definition

2.1 Vector addition

$$\vec{u} + \vec{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix} \quad \text{Given that: } \vec{u}, \vec{v} \in \mathbb{R}^n$$

Note: Vector addition is *commutative* meaning $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

2.2 Scalar multiplication

$$c\vec{v} = \begin{pmatrix} c_1v_1 \\ c_2v_2 \\ \vdots \\ c_nv_n \end{pmatrix} \quad \text{Given that: } c_1, c_2, \dots, c_n \in \mathbb{R}, \quad v_1, v_2, \dots, v_n \in \mathbb{R}^n$$

Scalar multiplication meaning:

1. If $c > 1$: Positive stretching
2. If $0 < c < 1$: Diminishing/shrinking vector

3. If $c < 0$: Flipping
4. If $c = 0$: Collapsing to a zero vector

2.3 Linear combination

Scaling vectors and adding them together.

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \cdots + c_n \vec{v}_n$$

Where: $c_i \in \mathbb{R}$ and $\vec{v}_i \in \mathbb{R}^n$

2.4 Span

Span is a set of all linear combinations of a set of vectors.

$$\text{span}\{\vec{v}_1 + \vec{v}_2\} = \text{span}\{c_1 \vec{v}_1 + c_2 \vec{v}_2 \mid c_i \in \mathbb{R}, \vec{v}_i \in \mathbb{R}^n\}$$

Geometry explanation:

1. Span of a non zero vector is a line through the origin.
2. Span of a two non-parallel vectors, spans the entire $\mathbb{R}^2$ space.
3. Span of two parallel vectors is a line in $\mathbb{R}^2$

3 Python Results

In Python, vectors are defined using arrays. Computing vector addition is as simple as adding the vectors together. Likewise, scalar multiplication is implemented by taking the scalar and multiply it with the array already defined. In Python multiplication is done using asterisk symbol (*).

When computing span of a vector, we use rank to get the total number of independent directions. Computing the magnitude of a vector is also possible and easy in Python. Magnitude is important when normalizing features e.g. in a feature matrix.

4 ML Connection

1. Adding features in a feature matrix
2. Normalizing features
3. Presenting predictions from a linear regression model. Linear regression predictions is a linear combination of weights and a target vector
4. Span of a feature matrix should be the same as the one for target vector. If the target vector lies outside the span, no exact solution exists.

5 Reflection

Vectors are essential in ML. They are used to represent some data.